

Low-Complexity Beam Selection Algorithms for Millimeter Wave Beam-space MIMO Systems

Rahul Pal¹, A. Krishna Chaitanya¹, and K. V. Srinivas¹

Abstract—We consider the downlink multi-user multiple-input multiple-output (MIMO) systems, operating at millimeter-wave frequencies and equipped with a large number of antenna elements at the access point. Beam-space MIMO with beam selection offers an attractive solution for reducing the number of required radio frequency chains in such systems. We propose two beam selection algorithms. The first one is a heuristic greedy algorithm that enjoys low computational complexity with moderate performance. By modeling beam selection as maximum weight matching over a bipartite graph, we propose beam selection based on the Kuhn–Munkres algorithm and, importantly, introduce a new formal framework for beam selection.

Index Terms—Precoding, mmWave communications, high dimensional multiuser-MIMO, beam-space, beam selection.

I. INTRODUCTION

COMMUNICATION at millimeter wave (mmWave) frequencies (30 GHz to 300 GHz) is going to be one of the enabling technologies in the emerging IMT 2020 (aka 5G) wireless systems [1], [2]. In addition to the availability of huge bandwidths, mmWave frequencies enable us to employ many antenna elements within a given aperture area, enabling *massive*, or, *high-dimensional* multiple input multiple output (MIMO) operation [3]. However, as each antenna element requires a separate radio frequency (RF) chain, it results in a huge hardware complexity. At mmWave frequencies, an RF chain consumes significantly higher power when compared to sub-6 GHz frequencies [2], [3]. Further, RF chain is a costly component of a communication system, making it highly difficult to afford a large number of RF chains.

The propagation of mmWaves can be made highly directional, which makes the mmWave channels *sparse* in the angular domain. Beam-space MIMO (B-MIMO), where the channel is transformed into angular domain from spatial domain [4]–[6], will allow us to exploit the inherent sparsity in the mmWave channels for decreasing the number of RF chains. Each beam in B-MIMO requires a single RF chain, and, by a careful selection of a small number of beams, we can reduce the number of RF chains, while *not* sacrificing much in the sum rate performance.

The concept of beam selection in mmWave B-MIMO has been proposed in [7]. The maximum magnitude criteria for

beam selection, adapted in [7], makes the number of simultaneously active RF chains a function of the channel realization. The interference-aware (IA) beam selection alleviates this drawback and achieves higher sum rate by reducing the interference among the users [8]. Amadori and Masouros [9] have proposed multiple algorithms for beam selection. An iterative beam selection, based on QR decomposition of the beam-space channel matrix, along with a new precoder, has been proposed in [10]; Though it achieves superior performance than other methods, it suffers from high computational complexity.

This letter presents two novel beam selection algorithms. Its main contributions are the following:

- Presents a greedy beam selection algorithm having very low computational complexity.
- Formulates beam selection as a *maximum weight matching* (MWM) problem over a bipartite graph, providing a novel formal framework for beam selection.

Notation: Bold uppercase letters denote matrices and bold lowercase letters denote vectors. a_{ij} is the $(i, j)^{\text{th}}$ entry of $\mathbf{A}$ and a_i is the i^{th} entry of $\mathbf{a}$. $\|\mathbf{A}\|$ is the Frobenius norm, $\mathbf{A}^{-1}$ is the inverse, $\mathbf{A}^T$ is the transpose and $\mathbf{A}^H$ is the Hermitian transpose of $\mathbf{A}$, respectively. Empty set is denoted by $\emptyset$ and $\mathcal{A} \setminus \mathcal{B} = \{x : x \in \mathcal{A} \text{ and } x \notin \mathcal{B}\}$ denotes set difference. $\binom{n}{k}$ denotes “ n choose k ”.

II. SYSTEM MODEL

Downlink multi-user MIMO communication between an access point (AP), having a uniform linear array (ULA) with N elements, and K users, each one equipped with a single receive antenna, is considered here. The discrete-time input and output of such a system, in spatial domain, are related as

$$\mathbf{y} = \mathbf{H}^H \mathbf{F} \mathbf{s} + \mathbf{z}, \quad (1)$$

where $\mathbf{h}_k \in \mathbb{C}^{N \times 1}$, k^{th} column of $\mathbf{H} \in \mathbb{C}^{N \times K}$, contains channel gains from N element ULA to the k^{th} user. Data vector $\mathbf{s} \in \mathbb{C}^{K \times 1}$ is precoded with $\mathbf{F} \in \mathbb{C}^{N \times K}$ to cancel the multi-user interference and $E[\|\mathbf{F} \mathbf{s}\|^2] \leq \rho$ represents the transmit power constraint. z_k denotes the additive white Gaussian noise at user k and $z_k, i = 1, \dots, K$, are assumed to be independent and identically distributed (IID) with $z_i \sim \mathcal{CN}(0, \sigma^2)$.

Let $\alpha_k^{(\ell)}$ denote the ℓ^{th} path gain from the ULA to the receive antenna of user k , where $\ell = 0$ corresponds to the line-of sight (LoS) component, and $\ell = 1, \dots, L$, represent L non-line-of-sight (NLoS) components. Following Saleh-Valenzuela channel model [11], [12],

$$\mathbf{h}_k = \sqrt{\frac{N}{L+1}} \sum_{\ell=0}^L \alpha_k^{(\ell)} \mathbf{a}(\theta_k^{(\ell)}). \quad (2)$$

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Assuming that the distance between successive elements in the ULA is $\frac{\lambda}{2}$ meters, where λ is the carrier wavelength in meters, the array response vector corresponding to ℓ^{th} path is given by

$$\mathbf{a}(\theta_k^{(\ell)}) = \frac{1}{\sqrt{N}} \left[\exp(-j2\pi\theta_k^{(\ell)}i) \right]_{i \in \mathcal{I}(N)}, \quad (3)$$

where $\theta_k^{(\ell)} = 0.5 \sin \phi_k^{(\ell)}$ is the spatial frequency corresponding to angle of arrival of the ℓ^{th} component at k^{th} user, denoted by $\phi_k^{(\ell)}$. Further, $\phi_k^{(\ell)}$ is considered uniformly distributed in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. $\mathcal{I}(N)$ is an index set, given by $\mathcal{I}(N) = \{m - (N-1)/2, m = 0, 1, \dots, N-1\}$.

As mmWave channels are sparse in the angular domain, *beam-space*, i.e., *angular domain*, representation is a better choice for mmWave channels. The spatial domain and the angular domain are related through a *spatial discrete Fourier transform*, represented by the unitary matrix $\mathbf{U} \in \mathbb{C}^{N \times N}$. The i^{th} column of $\mathbf{U}$, for $i \in \mathcal{I}(N)$, is given by $\mathbf{a}(\theta_i = \frac{i}{N})$ [4]–[6]. Thus, $\mathbf{U}$ contains, as its columns, the array response vectors corresponding to uniformly spaced spatial angles.

As a discrete lens array (DLA) plays the role of a spatial DFT, the spatial channel can be transformed into the beam-space channel by employing a DLA at the transmitter [6]. The modified beam-space system equation would be

$$\mathbf{y} = \mathbf{H}_b^H \mathbf{F}_b \mathbf{s} + \mathbf{z}, \quad (4)$$

where $\mathbf{H}_b = \mathbf{U}^H \mathbf{H} \in \mathbb{C}^{N \times K}$ is the beam-space channel and $\mathbf{F}_b = \mathbf{U}^H \mathbf{F} \in \mathbb{C}^{N \times K}$. The number of significant entries in each $\mathbf{h}_{b,k} = \mathbf{U}^H \mathbf{h}_k$, $k = 1, \dots, K$, is considerably less than N , representing the sparse nature of the mmWave channel.

Each beam in the beam-space channel (i.e., $\mathbf{H}_b$) needs an RF chain. Using only a few beams, out of N , reduces the required number of RF chains; As discussed in [7], the challenge is to select the beams *without* causing considerable performance loss. We address the problem of selecting K beams (out of N) with the sum rate $R_{\text{sum}} = \sum_k R_k$ as the performance metric. Here, $R_k = \log_2(1 + \text{SINR}_k)$ is the rate achieved by user k and SINR_k is the signal-to-interference plus noise ratio at user k [10]. The input-output relation of the system, after beam selection, is given by

$$\mathbf{y} = \tilde{\mathbf{H}}_b^H \tilde{\mathbf{F}}_b \mathbf{s} + \mathbf{z}, \quad (5)$$

where, $\tilde{\mathbf{H}}_b^H \in \mathbb{C}^{K \times K}$ is the *effective* channel matrix after beam selection and $\tilde{\mathbf{F}}_b \in \mathbb{C}^{K \times K}$ is the corresponding precoder to cancel the multi-user interference over the effective channel.

III. GREEDY BEAM SELECTION ALGORITHM

We now propose a greedy beam selection algorithm that allocates a beam for each user. Let $\mathbf{G} \in \mathbb{R}^{K \times N}$ be the channel gain matrix, where g_{kn} is given by the squared absolute value of the $(k, n)^{\text{th}}$ element of $\mathbf{H}_b^H$. Beam n is considered stronger to user k_1 than user k_2 , if $g_{k_1 n} > g_{k_2 n}$. Greedy beam selection, given by Algorithm 1, is a search based iterative procedure. Let β_k be the index of the strongest beam for user k . In the first iteration, we find $\beta_k = \arg \max_{n \in \{1, \dots, N\}} g_{kn}$, $k = 1, \dots, K$. If $\beta_k \neq \beta_{k'}$ for $k' \in \{1, \dots, K\} \setminus \{k\}$, i.e., if beam β_k is the strongest beam *only* for user k , then β_k is allocated to user k .

Algorithm 1 Greedy Algorithm for Beam Selection

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Input:  $\mathbf{G}_{K \times N}$ 
Initialize:  $\mathcal{N} = \{1, \dots, N\}$ ,  $\mathcal{K} = \{1, \dots, K\}$ 
for  $k = 1 \rightarrow K$  do
     $\beta_k = \arg \max_{n \in \{1, \dots, N\}} \{g_{kn}\}$ 
end for
ITER:
for  $k = 1 \rightarrow K$  do
    if  $k \in \mathcal{K}$  then
        if  $\beta_k \neq \beta_{k'}$  for  $k' \in \mathcal{K} \setminus \{k\}$  then
            Allocate beam  $\beta_k$  to user  $k$ 
             $\mathcal{K} \leftarrow \mathcal{K} \setminus \{k\}$  and  $\mathcal{N} \leftarrow \mathcal{N} \setminus \{\beta_k\}$ 
        else
            Allocate  $\beta_k$  to  $\kappa$  where,  $\kappa = \arg \max_{i \in \mathcal{K}} \{g_{i\beta_k}\}$ 
             $\mathcal{K} \leftarrow \mathcal{K} \setminus \{k\}$  and  $\mathcal{N} \leftarrow \mathcal{N} \setminus \{\beta_k\}$ 
        end if
    end if
end for
if  $\mathcal{K} \neq \emptyset$  then
     $\beta_k = \arg \max_{n \in \mathcal{N}} \{g_{kn}\}$ ,  $k \in \mathcal{K}$ 
    Goto ITER
else
    Stop
end if

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If a beam is the strongest beam for more than one user, then that beam will be allocated to the user having a higher channel gain. In other words, if $\beta_k = \beta_{k'}$ for $k \neq k'$, then β_k is allocated to k if $g_{kn} > g_{k'n}$; else, it is allocated to k' . For the remaining users, we repeat the process after removing the allocated beams from the set of available beams. This process is repeated until a beam is selected for each user. Note that Algorithm 1 finds a beam for each user, provided $N > K$.

The number of iterations in the greedy algorithm depends on the probability that a given beam is strongest to more than one user. By computing the probability that two or more users share a strongest beam, we show that Algorithm 1 requires less number of iterations if $K \ll N$. Assume that g_{kn} are IID and let $F_g(x)$ and $f_g(x)$ denote, respectively, the cumulative distribution function and probability density function of g_{kn} , $\forall k, n$. Without loss of generality, we consider user k' and beam n' and compute the probability, $p_{k'n'}$, that n' is the strongest beam for k' .

$$\begin{aligned} p_{k'n'} &= \Pr \{g_{k'n'} > g_{k'n}, \forall n \neq n'\} \\ &= \int_0^\infty (\Pr \{g_{k'n} < x \text{ for some } n\})^{N-1} dF_g(x) \\ &= \frac{1}{N}. \end{aligned} \quad (6)$$

Thus, $\frac{1}{N}$ is the probability that a given beam is strongest to a given user. The probability p that a given beam, out of N , is strongest to two or more users is given by

$$\begin{aligned} p &= \sum_{i=2}^K \binom{K}{i} \left(\frac{1}{N}\right)^i \left(1 - \frac{1}{N}\right)^{K-i} \\ &= 1 - \left(1 - \frac{1}{N}\right)^K - \frac{K}{N} \left(1 - \frac{1}{N}\right)^{K-1} \end{aligned} \quad (7)$$

For a fixed value of K , $p \rightarrow 0$ as $N \rightarrow \infty$. Thus, if N is large compared to K , the probability of two or more users having the same beam as their strongest beam is *small*. We call such a system, a *sparse* system. $p \rightarrow 1$ if $N \rightarrow \infty$ with $\frac{K}{N} \approx 1$; i.e., if the values of K and N are close to each other, with *high* probability, there exists a beam that is stronger to more than one user. We refer to such a system as a *dense* system. The greedy algorithm requires less number of iterations in a sparse system when compared to a dense system.

IV. BEAM SELECTION BASED ON MAXIMUM WEIGHT MATCHING (MWM)

We model the beam selection problem, of selecting K beams out of N , as a maximum weight matching problem over a bipartite graph. Let $B = \{b_1, \dots, b_N\}$ be the set of beams and let $U = \{u_1, \dots, u_K\}$ be the set of users. Consider a bipartite graph $G = (V, E)$ where $V = B \cup U$ and $(b_n, u_k) \in E$ for $1 \leq n \leq N$, $1 \leq k \leq K$. We assign $w_{nk} = \log_2(1 + \text{SINR}_{nk})$ as the weight of the edge (b_n, u_k) , where,

$$\text{SINR}_{nk} = \frac{S_{nk}}{I_k + \sigma^2}. \quad (8)$$

Here, S_{nk} is the signal power at user k due to beam n , I_k is the interference power at user k from *all* the beams *except* beam n . A matching over G is defined as a set M , where $M \subseteq E$, which satisfies the following:

- 1) for each $k \in \{1, \dots, K\}$, $\exists n \in \{1, \dots, N\}$ such that $(b_n, u_k) \in M$, i.e., each user will get *at least* one beam,
- 2) for each $n \in \{1, \dots, N\}$, there exists *at most* one $k \in \{1, \dots, K\}$ such that $(b_n, u_k) \in M$, i.e., a beam is allocated to *at most* one user.

The problem of MWM is to find a matching M^* that maximizes the sum of edge weights in the matching, which can also be stated as the following optimization problem:

$$\begin{aligned} & \text{maximize} \sum_{k,n} w_{nk} x_{nk} \\ & \text{subject to} \sum_n x_{nk} = 1, \quad k = 1, \dots, K, \\ & \sum_k x_{nk} \leq 1, \quad n = 1, \dots, N, \\ & x_{nk} \in \{0, 1\}. \end{aligned}$$

Note that the optimization variables are x_{nk} , $1 \leq n \leq N$, $1 \leq k \leq K$. Kuhn-Munkres algorithm [14], [15], solves the MWM problem in an efficient way for $N = K$. We employ the Kuhn-Munkres algorithm modified for the general case of $N \neq K$ in [16]. Kuhn-Munkres algorithm has a complexity of $\mathcal{O}(NK^2)$ [16]. While the greedy algorithm is a much simpler solution, it requires more iterations when the values of N and K becomes comparable to each other.

While computing SINRs using (8), we have considered that *all* the N beams are active. This results in over estimation of the interference as only K beams are activated at any given time to serve K users. More importantly, interference seen by a user is determined by the beams assigned to the remaining $K - 1$ users, which implies that, when a user

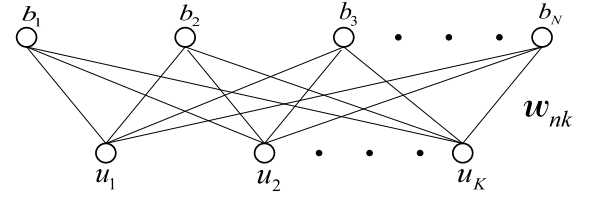


Fig. 1. Beam selection as MWM over a bipartite graph.

is assigned a beam (i.e., when an edge is selected), all the remaining edge weights change. When the MWM finds a matching, the SINRs will change and it is possible, with the modified edge weights, that a new set of edges maximize the sum rate. This *inter-dependency* among the edge weights makes the MWM based beam selection, presented here, *sub-optimal*. At the same time, this method achieves *comparable* performance, at significantly low complexity, relative to the existing beam selection methods, as shown in the next section.

V. PRECODING

Precoding, based on the effective channel matrix after beam selection, $\tilde{\mathbf{H}}_b$, is employed to eliminate the interference among the users. Zero-forcing (ZF) is a popular precoding technique; However, ZF *degrades* the performance when $\tilde{\mathbf{H}}_b$ is ill-conditioned and it can *not* be applied when $\tilde{\mathbf{H}}_b$ is rank deficient. A precoder that diagonalizes $\tilde{\mathbf{H}}_b$, based on QR decomposition of $\tilde{\mathbf{H}}_b$, has been proposed in [10]. We refer to this later precoder as “QR-Pr” and employ the same for canceling the interference. With QR-Pr precoder, the received signal at k^{th} user is given by,

$$y_k = \tilde{r}_{kk} s_k + z_k, \quad k = 1, \dots, K, \quad (9)$$

where $\tilde{r}_{kk}$ is the k^{th} diagonal element of $\tilde{\mathbf{R}}$, where $\tilde{\mathbf{H}}_b = \tilde{\mathbf{Q}}\tilde{\mathbf{R}}$.

VI. POWER ALLOCATION

As discussed above, QR-Pr creates a set of K *parallel* channels, with $\tilde{r}_{kk}$ as the gain of k^{th} channel. Assuming Gaussian signaling, it is well known that *water-filling* maximizes the sum rate across parallel channels [17]. Thus, instead of uniform power allocation, we propose to employ water-filling power allocation, given by,

$$P_k = \max \left\{ 0, \lambda - \frac{1}{\tilde{r}_{kk}^2} \right\}, \quad k = 1, \dots, K, \quad (10)$$

where, λ is chosen such that $\sum_k P_k \leq \rho$. Note that, with ZF precoding, all the users will have the *same* channel gain and there is *no* scope of power allocation.

VII. NUMERICAL EXAMPLES

We present performance of proposed algorithms, obtained through Monte Carlo simulations, considering two different systems: A *sparse* system, with $N = 256$ and $K = 32$ and a *dense* system with $N = 64$ and $K = 32$. Each user has a single receive antenna. We consider one LoS component with $\alpha_k^{(0)} \sim \mathcal{CN}(0, 1)$ and two NLoS components with $\alpha_k^{(1)} \sim \mathcal{CN}(0, 10^{-2})$ and $\alpha_k^{(2)} \sim \mathcal{CN}(0, 10^{-3})$, between the antenna

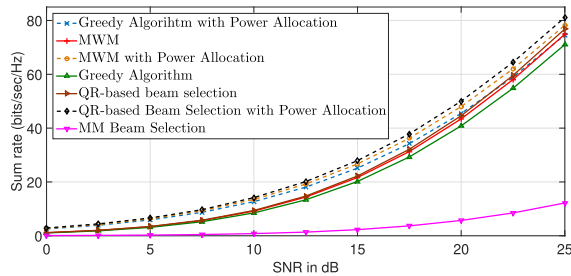


Fig. 2. Sum rate performance of different beam selection algorithms for a sparse system, with $N = 256$, and $K = 32$.

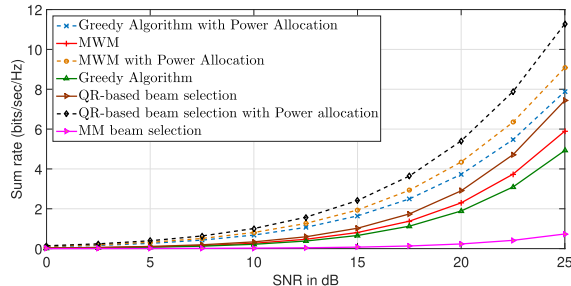


Fig. 3. Sum rate performance of different beam selection algorithms for a dense system, with $N = 64$ and $K = 32$.

array at AP and k^{th} user. $\alpha_k^{(\ell)}$ are mutually uncorrelated. Spatial frequencies, $\theta_k^{(\ell)}$, $\forall k, \ell$, are chosen uniformly randomly over $[-\frac{1}{2}, \frac{1}{2}]$ and are independent of each other.

Fig. 2 corresponds to the sparse system. “QR-based beam selection” refers to the algorithm proposed in [10] and shown to outperform all the existing algorithms including the IA beam selection [8] and the full dimensional ZF [7]. “MM beam selection” refers to the maximizing magnitude beam selection [7], [9], which is, by far, the *simplest* algorithm. Observe that the performance of MWM beam selection is *very close* to that of the QR-based beam selection. The QR-based beam selection is an iterative algorithm that computes QR decomposition of an $(N-i) \times K$ matrix in i^{th} iteration, $i = 0, \dots, (N-K-1)$, resulting in an overall complexity of $\sum_{i=0}^{N-K-1} (N-i) \mathcal{O}(2(N-i)K^2)$. With a complexity of only $\mathcal{O}(NK^2)$, the MWM beam selection becomes an attractive alternative: *without considerable loss in the sum rate performance, MWM saves significantly in computational complexity*. The greedy algorithm, which is a much simpler algorithm compared to MWM, results in a comparable sum rate performance. Thus, if we have stringent limitations on the affordable computational complexity, greedy algorithm provides us with a better choice for beam selection.

Fig. 3 shows that, in a dense system, the performance gap between the QR-based beam selection and the proposed algorithms is more noticeable. Thus, in a dense system, to enjoy the benefits of low complexity of the proposed algorithms, we have to accept noticeable loss in the sum rate. At the same time, it should be noted that AP can be equipped with a very large number of antenna elements at mmWave frequencies, making N to be very high; Further, as the radius of a cell in mmWave cellular systems is small (≈ 50 to 100 meters), the number of active users per cell (K), would also be small. Thus, we can expect to encounter sparse

systems having $N \gg K$ more often than dense systems in mmWave cellular communications, making the proposed algorithms to be useful in practice. In both the figures, solid lines correspond to uniform power allocation and dashed lines correspond to water-filling. Optimal power allocation improves sum rate significantly, particularly when the system is dense, having more interference.

VIII. CONCLUSION

We have proposed two beam selection algorithms for mmWave multi-user beamspace MIMO downlink systems. The iterative greedy algorithm selects beams focusing only on the signal strengths at the users. By modeling the beam selection as a maximum weight matching problem over a bipartite graph, we proposed another solution. Both the algorithms offer an efficient way of selecting beams, especially in sparse systems. Further investigation into “beam selection as a bipartite matching” might lead to more efficient and/or distributed solutions for beam selection.

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